

Igor Varejão

MASTER STUDENT FELLOW

COMPUTER SCIENCE DEPARTMENT - UNIVERSIDADE FEDERAL DE MINAS GERAIS - BRASIL

igor.varejao.dev@gmail.com

EDUCATION

Master of Science in Computer Science, Artificial Intelligence

Federal University of Minas Gerais

Research focus: AI Music Generation and Music Information Retrieval (MIR)

Belo Horizonte, MG, Brazil

August 2025 – Present

Bachelor in Computer Science

Federal University of Espírito Santo

Vitória, ES, Brazil

March 2019 – November 2024

INDUSTRY EXPERIENCE

• AI Engineer Fellow

Ministério Público de Minas Gerais

Fellowship

August 2025 – Present

- Engineered data processing pipelines to extract and parse complex HTML sources, generating structured datasets for downstream modeling. Developed data ingestion workflows for PostgreSQL databases. Currently developing a robust LangChain data extraction agent supporting multiple formats.

• Data Engineer at Olho Do Dono

Full-time

Vitória, ES, Brazil

October 2024 – May 2025

- Created and maintained CI/CD pipelines, using Azure, Docker, and PostgreSQL to support infrastructure and data processing. Scaled cattle weight prediction systems and optimized computer vision workflows for production.

• Data Engineer Intern at Olho Do Dono

Internship

Vitória, ES, Brazil

July 2022 – September 2024

- Developed video-based cattle weight prediction systems in Python. Led image annotation, segmentation, tagging, and data ingestion workflows in CVAT, with rigorous quality checks. Conducted EDA with Jupyter, Briefer, and Plotly, and coordinated servers through ZMQ.

TEACHING EXPERIENCE

Teaching Assistant – Supervised Machine Learning

Federal University of Minas Gerais, March 2026 – June 2026

- Supported instruction in KNN, linear and logistic regression, SVM, and decision trees; guided students implementing foundational models with NumPy and scikit-learn.

Teaching Assistant – Neural Networks and Deep Learning

Federal University of Minas Gerais, August 2025 – December 2025

- Supported instruction in neural network training, convolutional and recurrent networks; guided students through PyTorch implementations and deep learning best practices.

TECHNICAL EXPERTISE

- Programming Languages:** Python, C, C++, Java, Haskell, Shell Script
- Machine Learning and Scientific Computing:** PyTorch, PyTorch Lightning, scikit-learn, NumPy, Pandas, OpenCV, LangChain
- Data and Visualization:** PostgreSQL, MySQL, Plotly, Jupyter, Briefer
- Infrastructure and Development:** Docker, Azure, AWS, Git, CI/CD, Linux, ZMQ, CVAT

RESEARCH EXPERIENCE

Artificial Intelligence Undergraduate Researcher

Federal University of Espírito Santo, August 2020 – July 2024

- Developed deep neural networks for defect pattern recognition in submersible centrifugal pumping systems in partnership with Petrobras. Processed vibration signals across time, frequency, and time-frequency domains; performed EDA and statistical testing; and developed the public VibData framework for centralized research datasets.

Artificial Intelligence Volunteer Researcher

Federal University of Espírito Santo, August 2024 – December 2024

- Developed transfer-learning models for rotary machinery fault diagnosis with the University of Ottawa. Built a benchmarking framework to evaluate pretrained models on centralized bearing datasets against state-of-the-art approaches.

SELECTED PUBLICATIONS

- I. M. dos Santos Varejão, L. G. de Oliveira Costa, L. H. P. da Silva, A. Rodrigues, M. P. Ribeiro, F. M. Varejão, and T. Oliveira-Santos, “The similarity bias problem: What it is and how it impacts vibration based intelligent fault diagnosis,” *Mechanical Systems and Signal Processing*, vol. 235, p. 112822, 2025.
- M. Sehri, I. Varejão, Z. Hua, V. Bonella, A. Santos, F. de Assis Boldt, P. Dumond, and F. M. Varejão, “Towards a universal vibration analysis dataset: A framework for transfer learning in predictive maintenance and structural health monitoring,” 2025.
- I. C. Silva, R. M. Salgado, I. M. S. V. ao, and F. M. V. ao, “Analysis and improvement of machine learning models for detecting street lighting lamps,” *Learning & Nonlinear Models*, vol. 21, no. 2, pp. 16–28, 2023.

FURTHER INFORMATION

• igorvarejao.tech • linkedin.com/in/igor-varejao • github.com/ivarejao